

The Life of a Bill:

Predicting Legislative Status in the Philippine Senate Through Machine Learning





The Philippine Senate is a highly selective legislative body composed of 24 nationally elected senators.



The legislative process

First Reading:

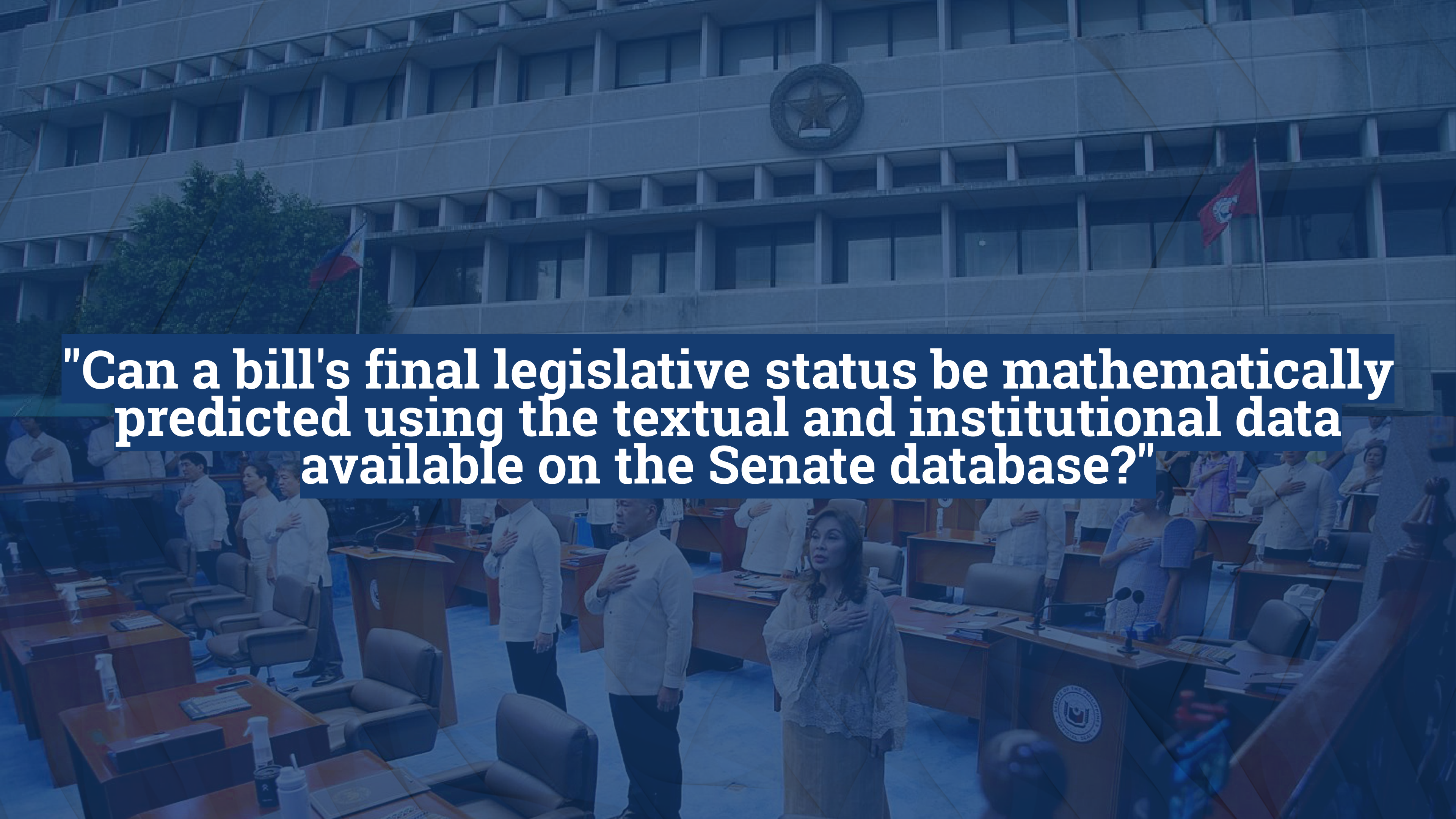
Bill is introduced and sent to a committee for review.

Second Reading:

Bill is debated, amended, and refined.

Approved on Third Reading:

Final vote is taken.

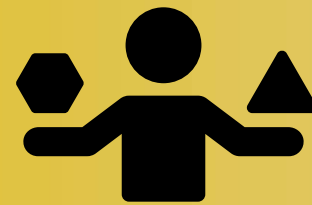


"Can a bill's final legislative status be mathematically predicted using the textual and institutional data available on the Senate database?"

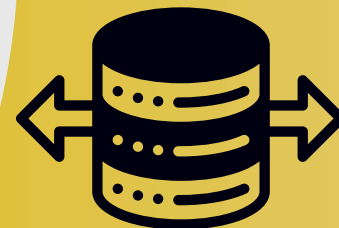
Methodology



Data Collection & Initial Processing



Exploratory Data Analysis (EDA)



Data Preparation & Feature Engineering



Model Building, Evaluation & Interpretation

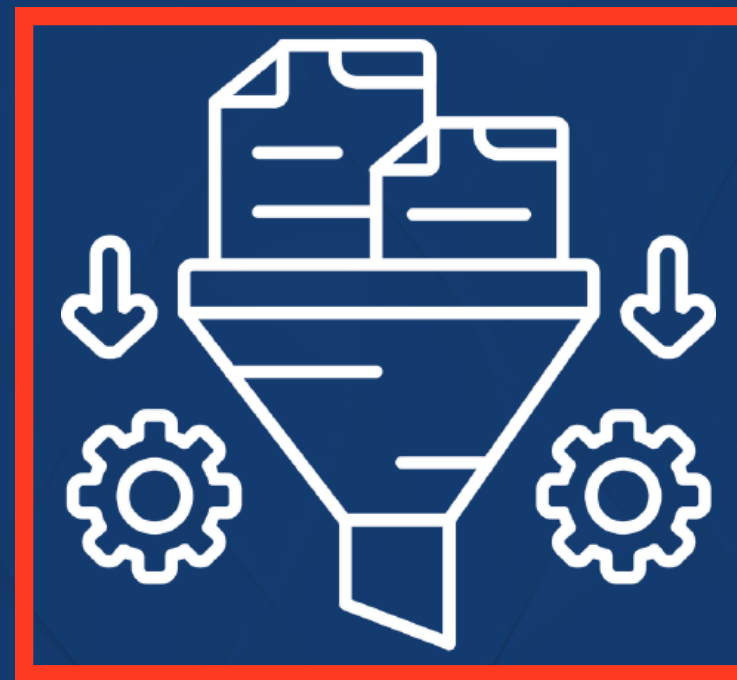
Data Collection & Initial Processing

Web Scraping & Data Source



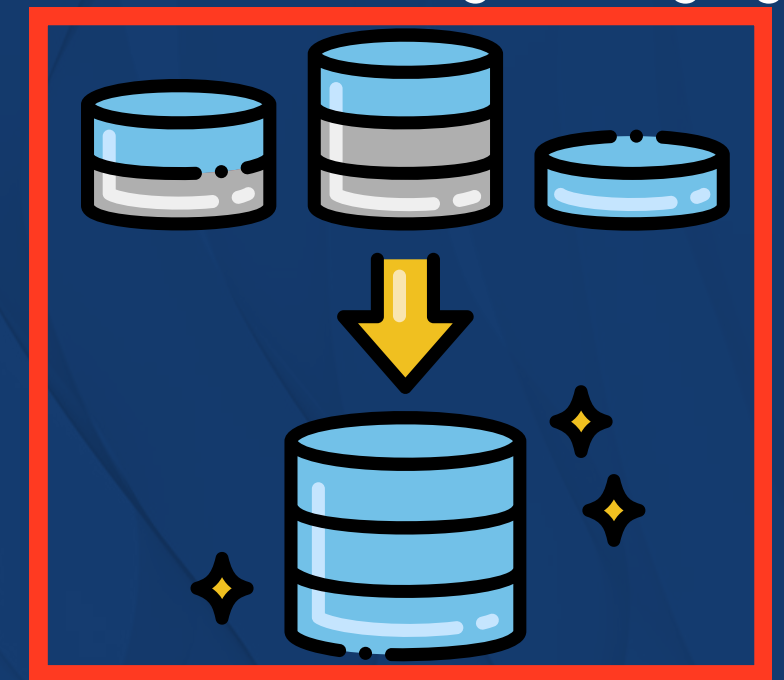
Philippines Legislative Document Room (LDR) database

Raw Data Extraction

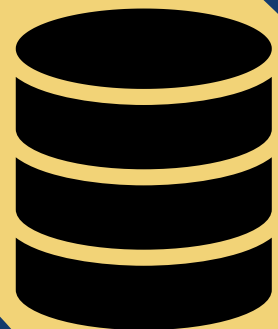


7,506 individual bills

Initial Processing & Merging



Removal of null rows AND merging of Legislative Records from the 15th-18th Congress





DATASET OVERVIEW

PHILIPPINES SENATE LEGISLATIVE DATASET

A comprehensive collection of senate bills from the 15th to 18th Congress, curated to understand legislative patterns and predict outcomes.



7,352

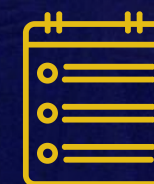
CLEANED BILLS

Cleaned and preprocessed
high-quality records



15th - 18th
CONGRESS

Spanning four congress for
broader insights



**12 SCRAPED
FEATURES**

Legislative features extracted
from Senate records



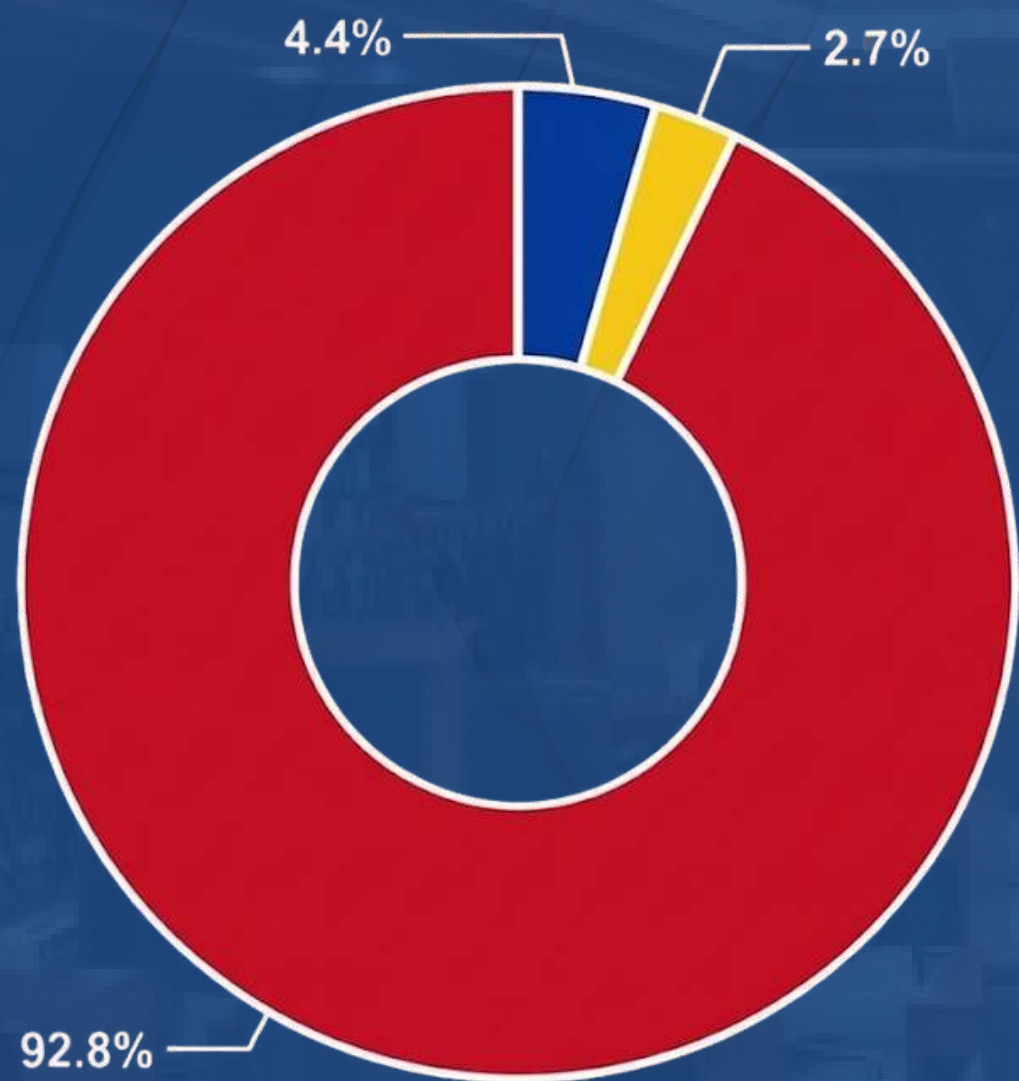
**SOURCE:
SENATE LDR**

Philippine Senate Legislative
Document Repository



THE LEGISLATIVE GRAVEYARD: SEVERE CLASS IMBALANCE IN THE DATASET

Proportional Share of Each Milestone



- First Reading (6,823 bills)
- Approved (327 bills)
- Second Reading (202 bills)

❗ 93% of bills never survive committee stage



Massive legislative bottleneck exists

Target Variable Consolidation

21 original Legislative
Statuses



Status Consolidation
& Mapping



First Reading
(Early Stage)

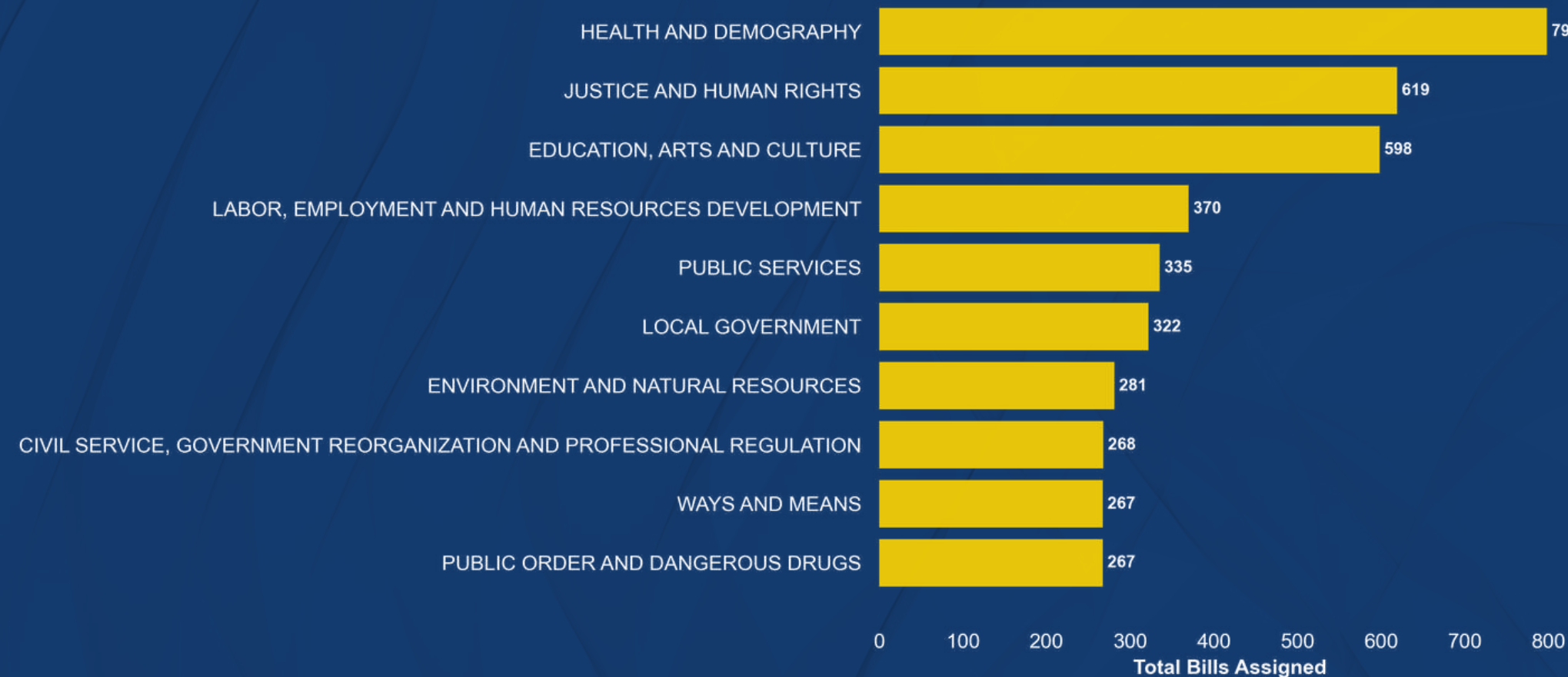


Second Reading
(Deliberation Stage)



Third Reading
(Final Stage)

Exploratory Data Analysis (EDA)

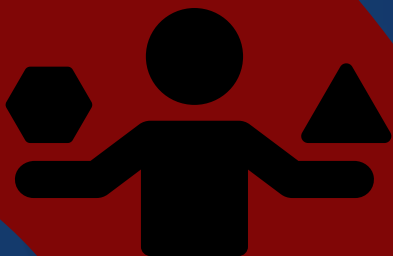
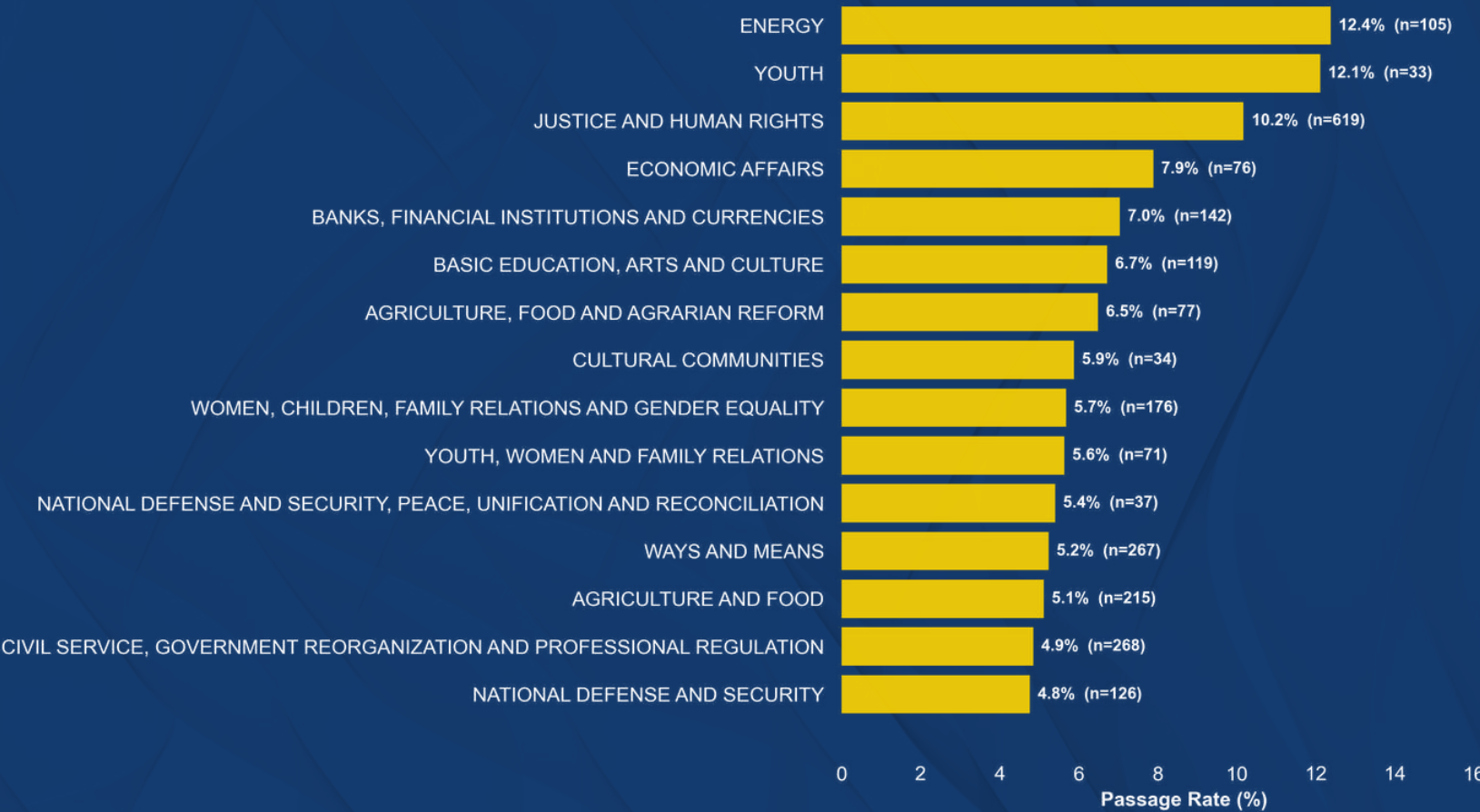


Bill Passage Rate by Senate Committee

Energy (12.4%) and Youth (12.1%) have the highest success rates, but Justice stands out: it passes 10.2% of a very large volume (619 bills). Most committees pass fewer than 7% of their bills.

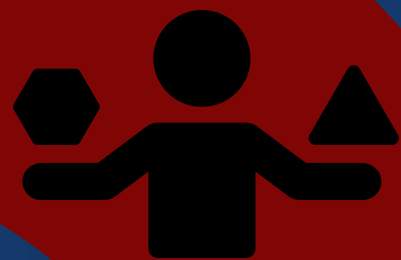
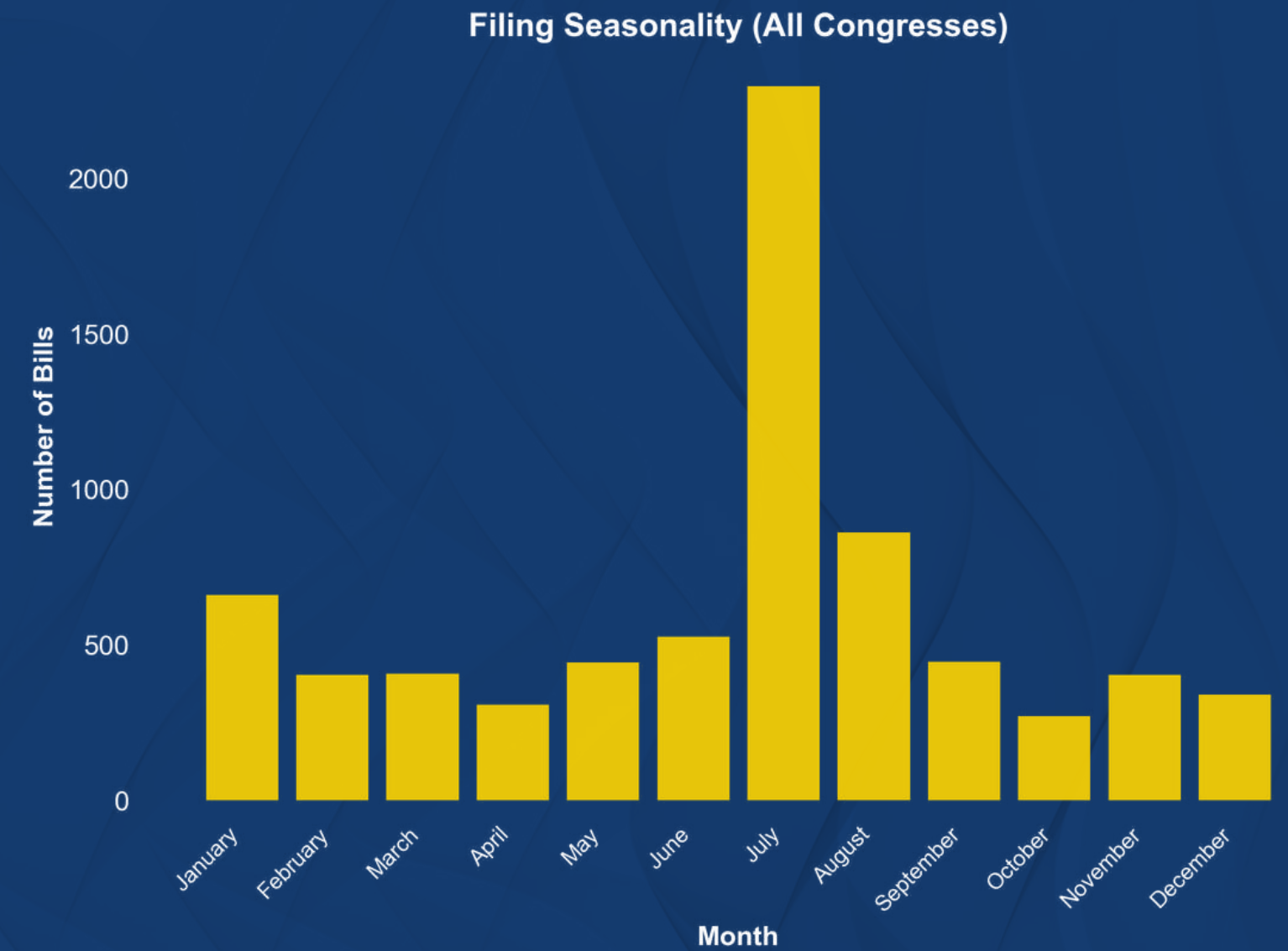
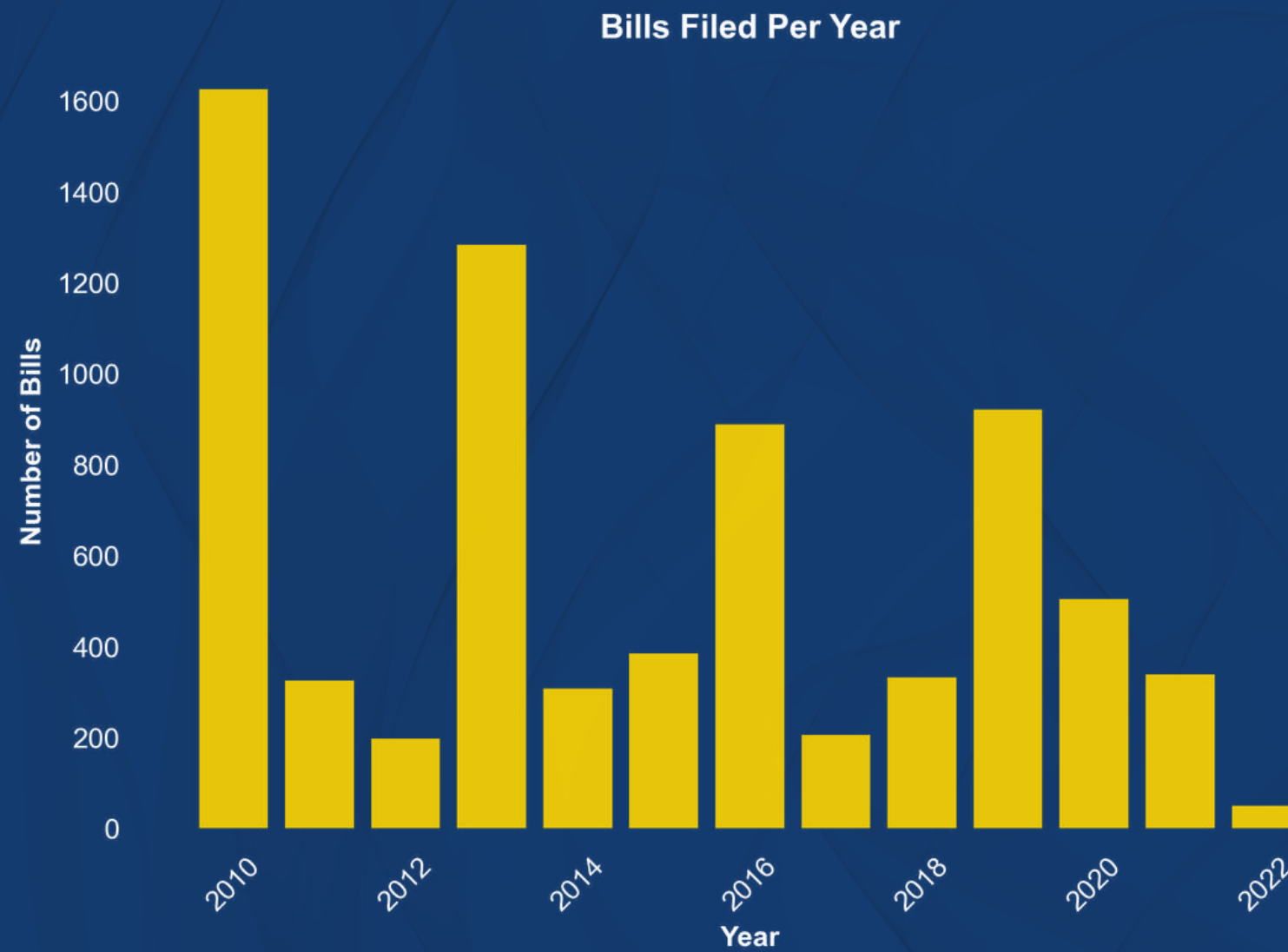
Top 10 Busiest Senate Committees

Health, Justice, and Education face the heaviest legislative bottlenecks, suggesting high public demand or complex policy volumes in those areas.



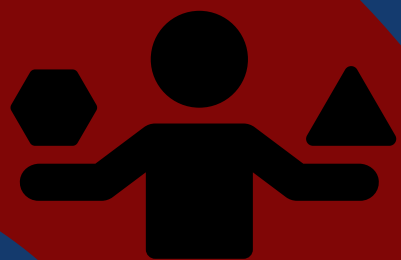
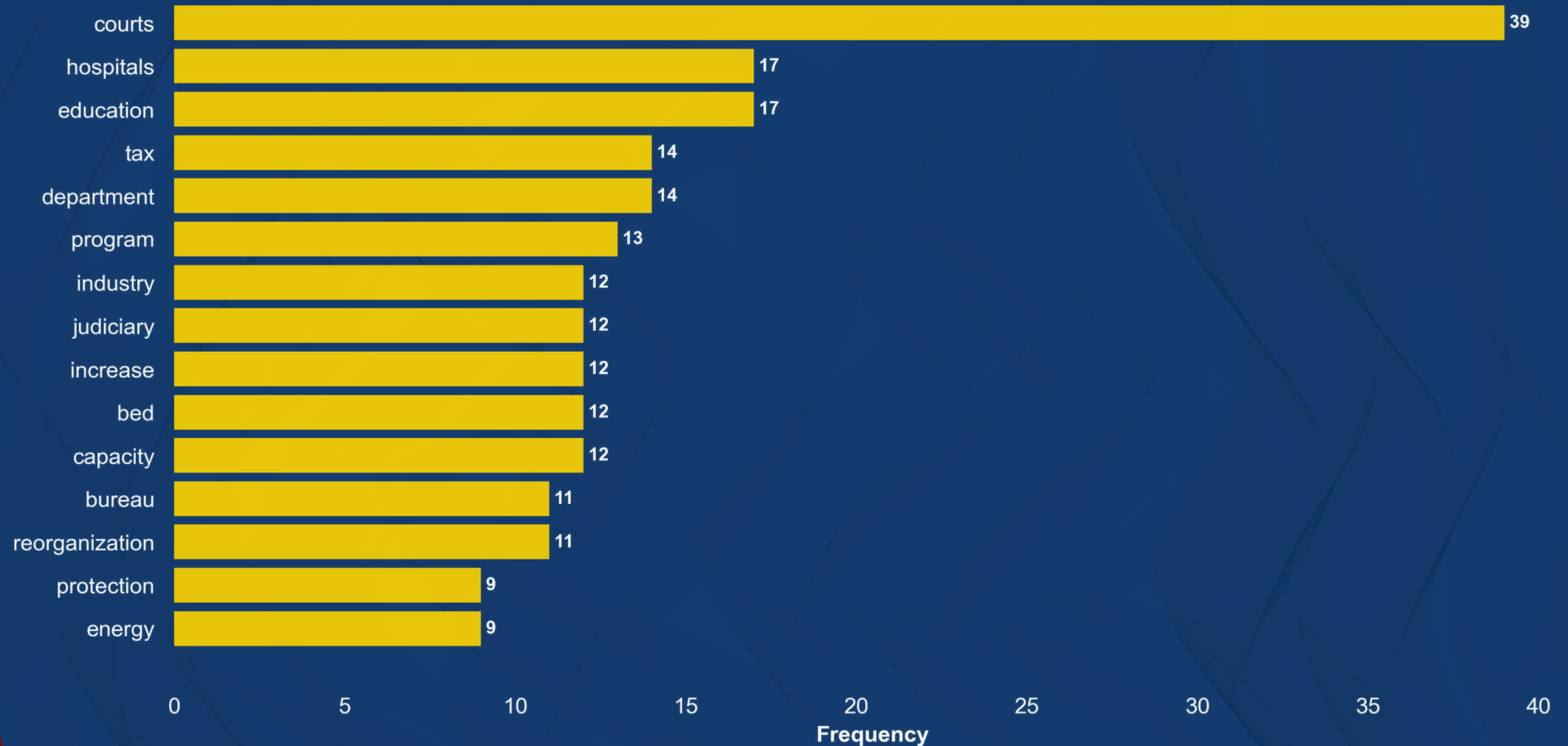
Exploratory Data Analysis (EDA)

Temporal Filing Patterns : When Are Bills Introduced?



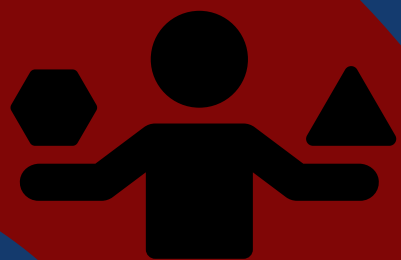
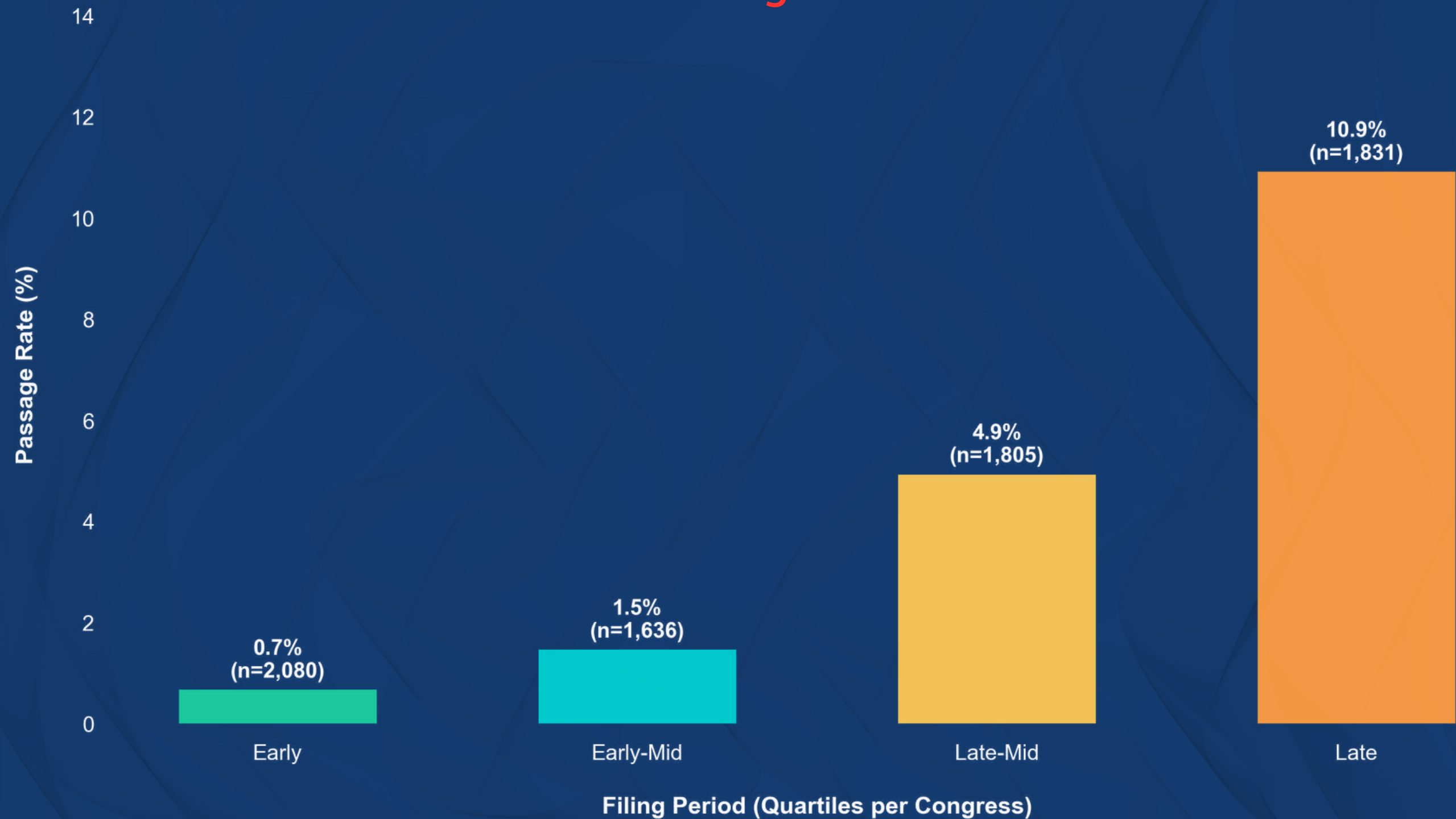
Exploratory Data Analysis (EDA)

Linguistic Signifiers



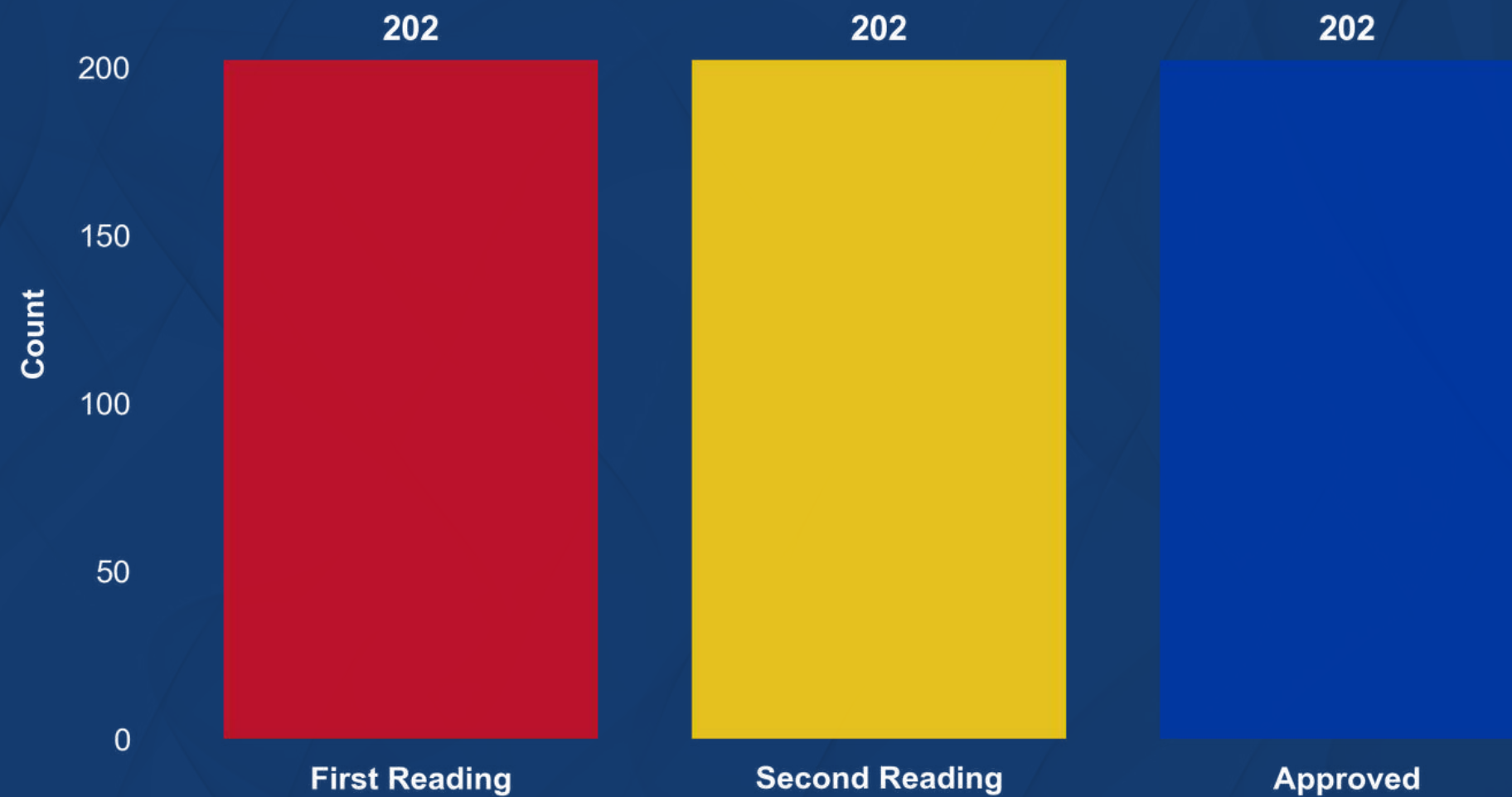
Exploratory Data Analysis (EDA)

Bill Passage Rate by Filing Period within Congressional Term



Feature Engineering

Undersample the majority class



Undersampling balances all three classes 1:1:1 by discarding excess "First Reading" rows. Equal representation forces the model to learn what actually predicts the rare "Approved" outcome.

Feature Engineering

01
10

**Indicator Variable
Creation
(scope)**



**Historical Efficacy
Encoding
(committee and
senators)**



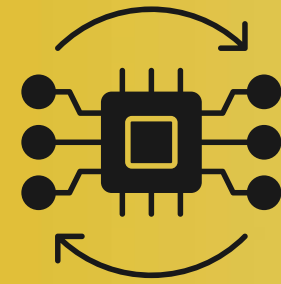
**Text Vectorization
(subject)**



**Temporal Feature
Engineering
(date filed)**



Data Modeling



Classification Algorithms



Feature Importance



Model Evaluation



Model Interpretation

Models Explored

LightGBM

XGBoost

RandomForest

NaiveBayes

LogisticRegression

k-NN

DecisionTree

LinearSVC

NeuralNet

Classification Algorithms

1 *F1:* 60.08%
Recall: 59.96%

2 *F1:* 59.16%
Recall: 59.09%

3 *F1:* 53.76%
Recall: 55.12%

 Baseline(PCC): 41.66%

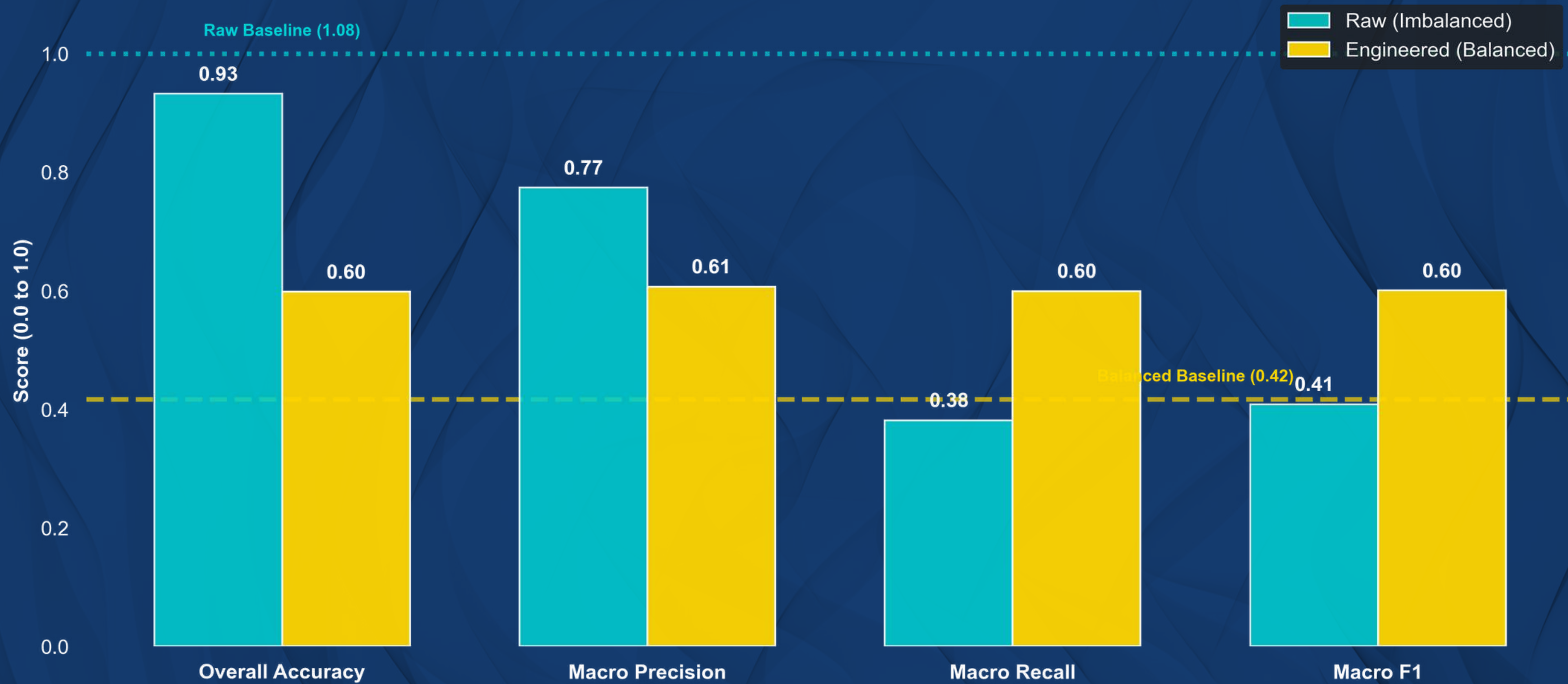
XGBoost
59.84%

LightGBM
59.02%

RandomForest
53.76%

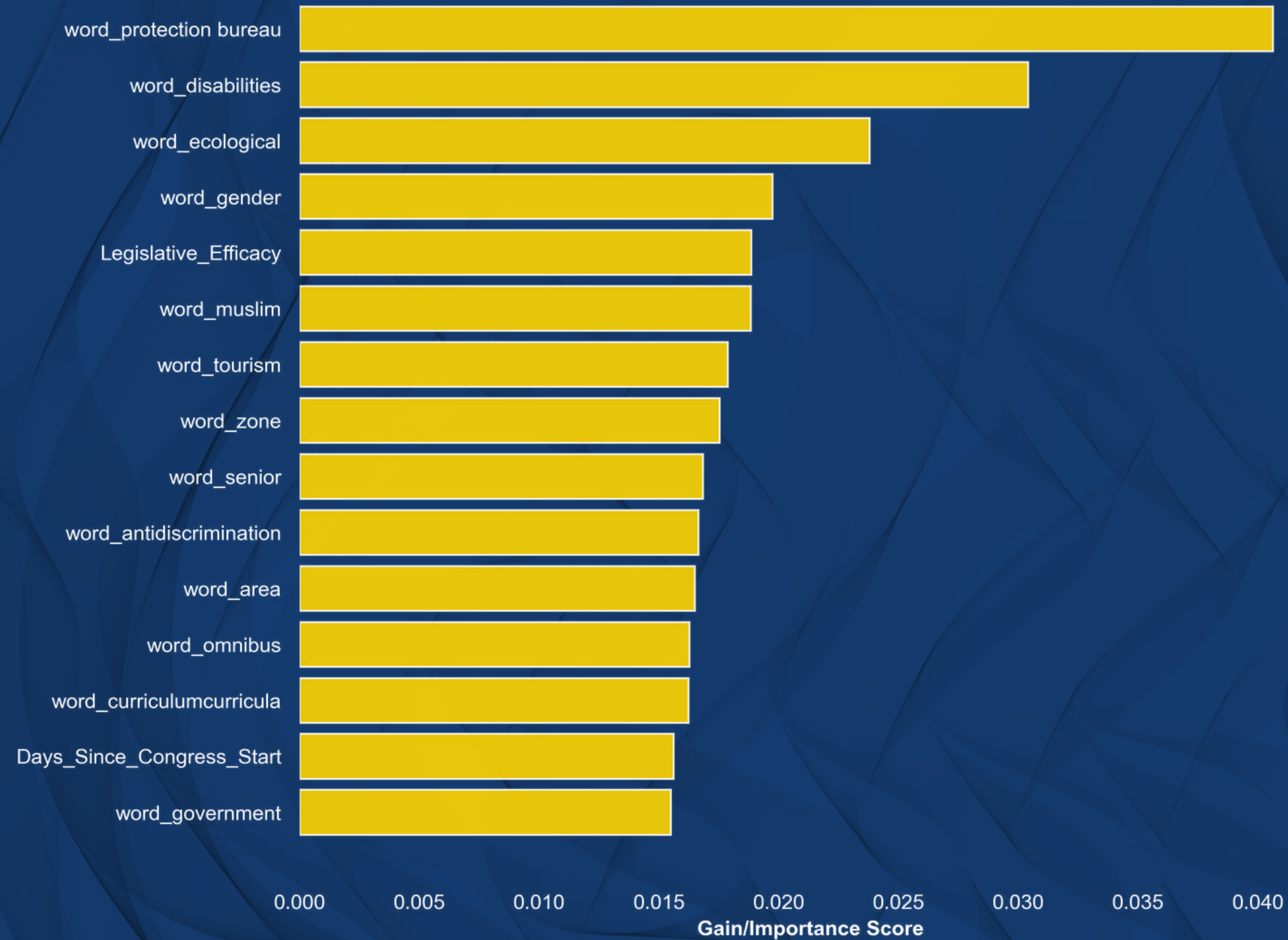


Comparison (Raw and Balanced)



FEATURE IMPORTANCE

Top 15 Most Important Features (XGBoost)



Conclusion

Top 3 Predictive Features



Protection Bureau



Disabilities



Ecological Policy

XGBoost

Achieved the highest predictive performance among all evaluated models

MODEL PERFORMANCE COMPARISON

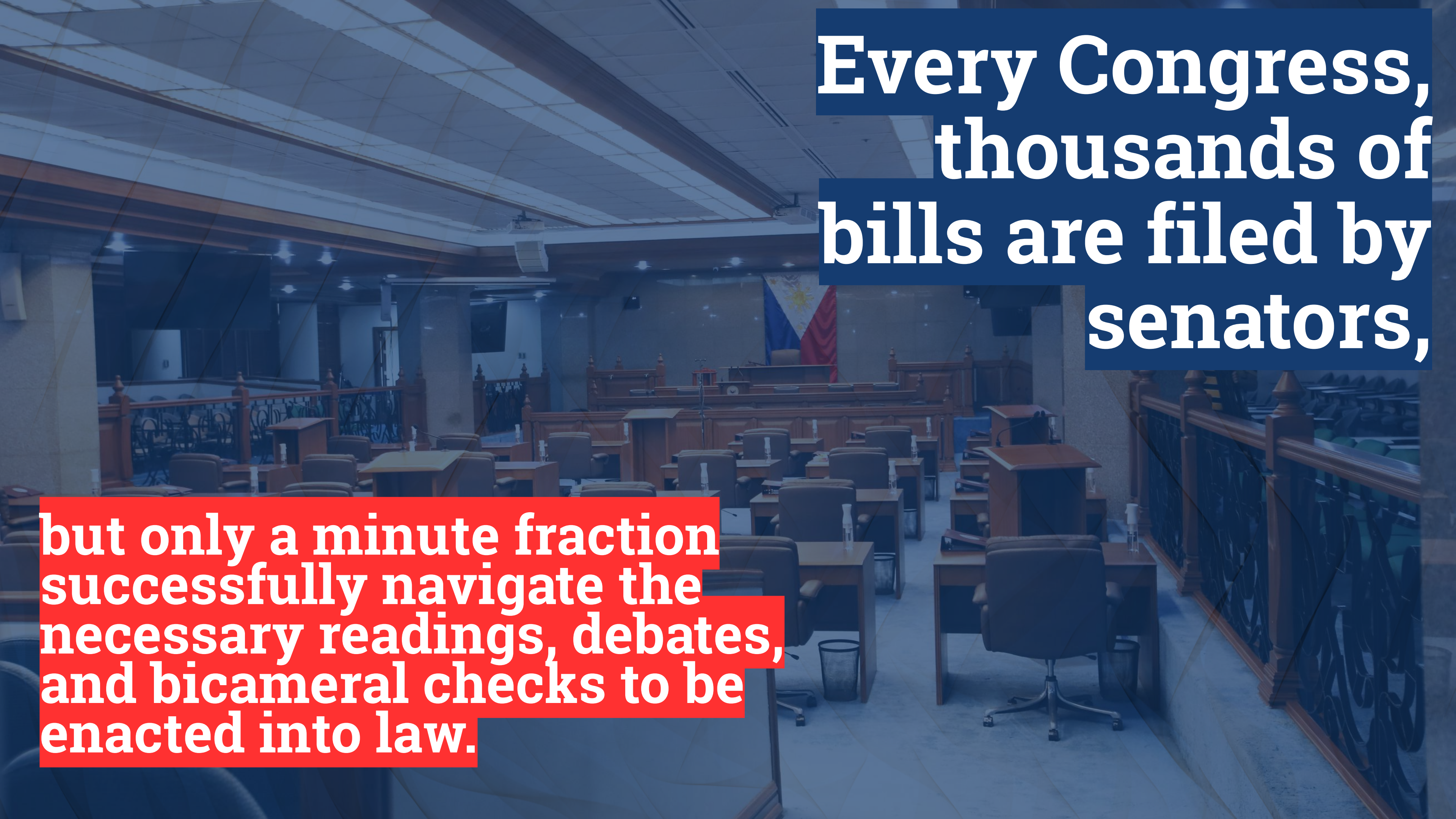


4.5%

The low legislative passage rate reflects a **highly selective institutional legislative filtering mechanism**

VOTE

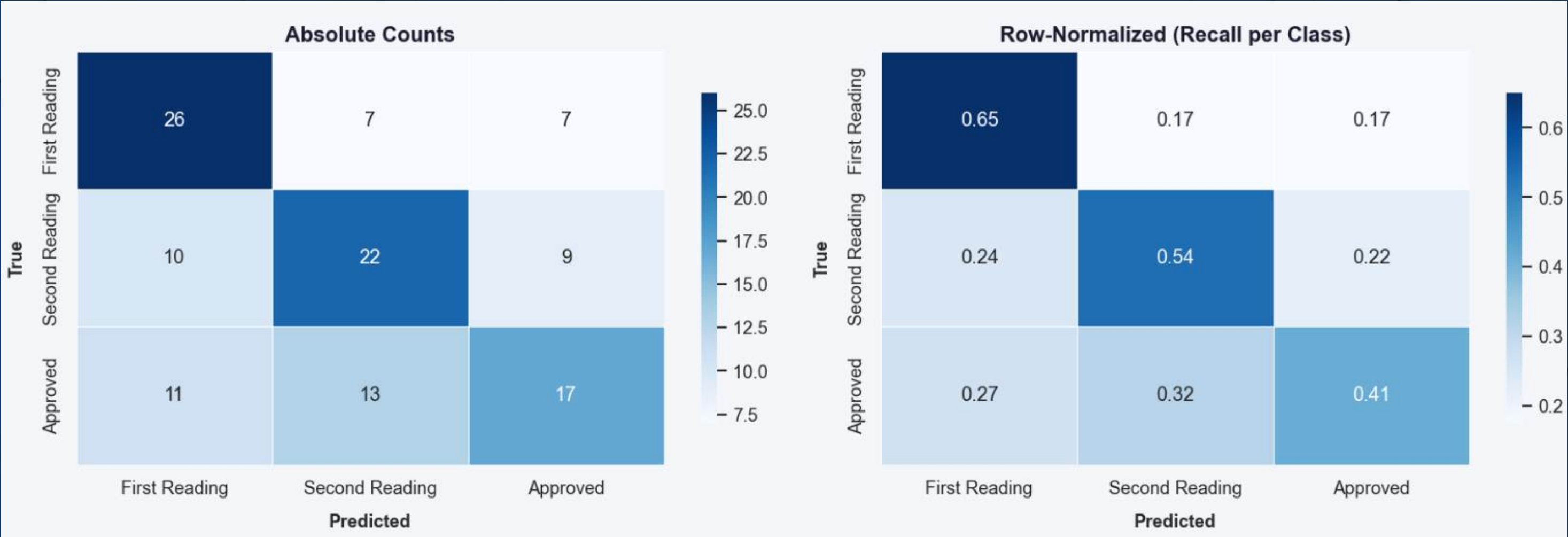
WISELY



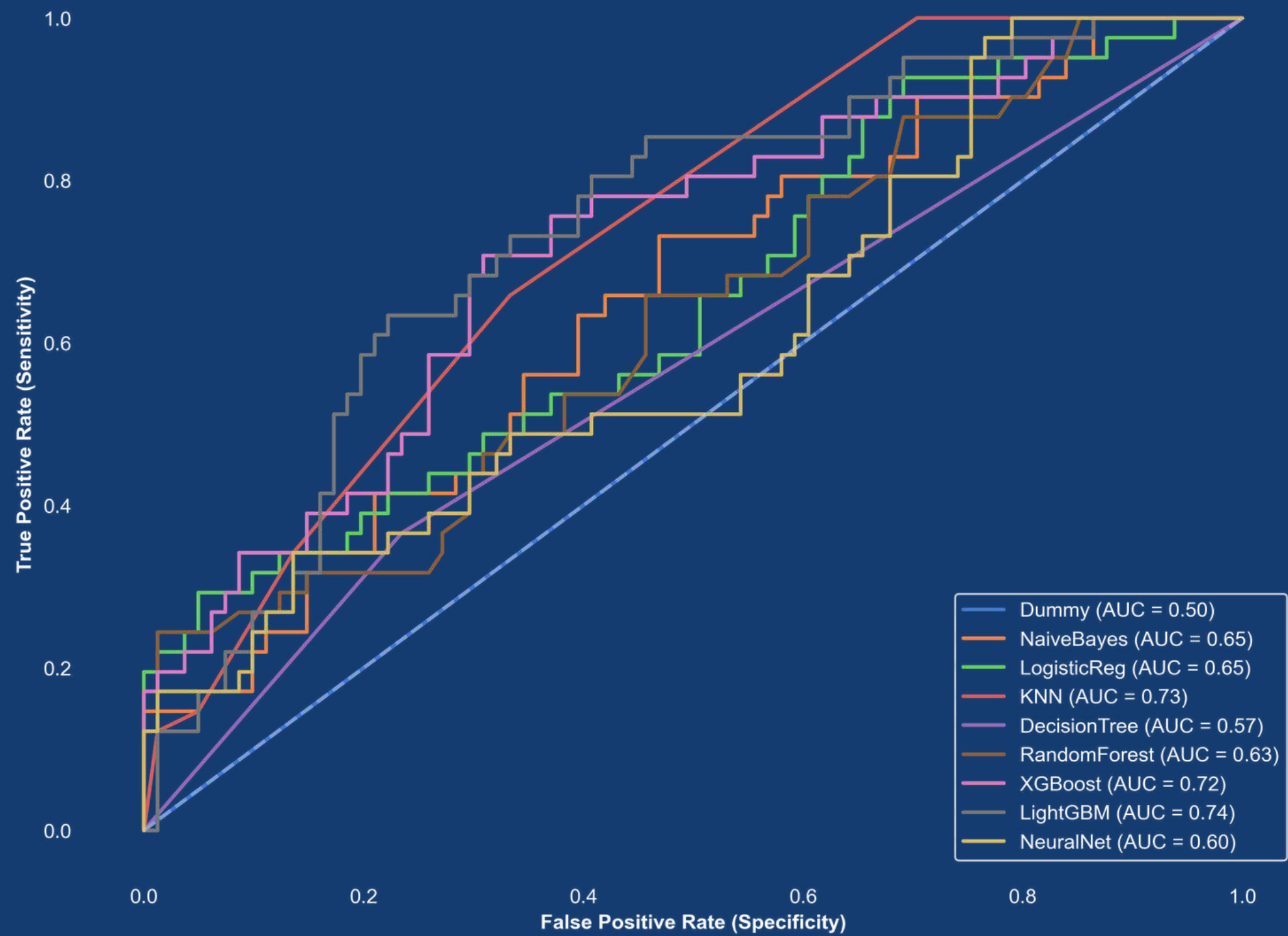
**Every Congress,
thousands of
bills are filed by
senators,**

**but only a minute fraction
successfully navigate the
necessary readings, debates,
and bicameral checks to be
enacted into law.**

confusion matrix



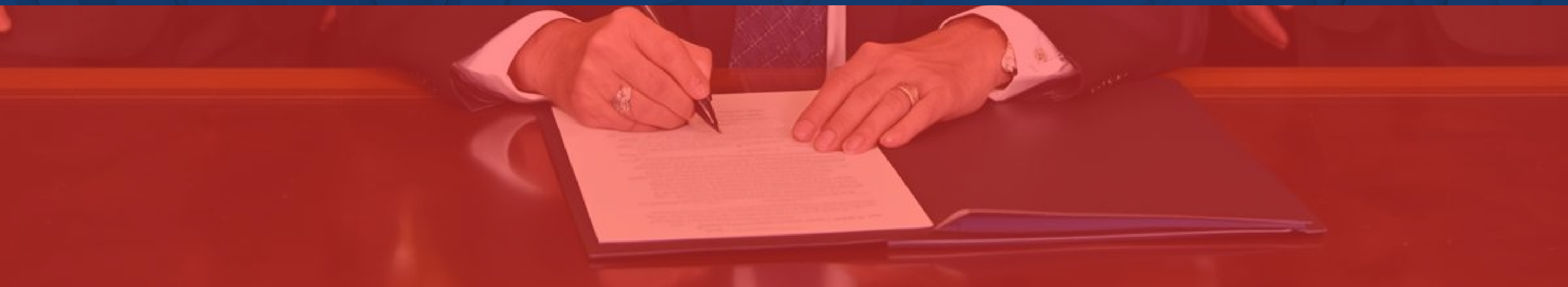
ROC-AUC Comparison: All Models (Target: Approved Class)



chuchu

chchu statement

chuchu



Historical data from the **15th to 18th** Congresses reveals

A massive **“Legislative Graveyard,”**
where most bills remain stuck in
the First Reading stage.

